LearnPath

AI-Powered Personalised Learning Platform

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1.0 Introduction

1.1 Background of the Case Study

Access to quality education remains one of the most pressing challenges across Southeast Asia. Despite significant advances in internet connectivity, millions of students in the ASEAN region continue to face barriers rooted in language, economic inequality, and the lack of personalised learning support. In countries such as Malaysia, Vietnam, and the Philippines, students frequently navigate educational systems that offer limited individual attention, rely on rote memorisation, and provide scarce academic resources outside of formal school hours.

LearnPath was conceived to directly address these systemic gaps. By combining the power of modern AI — specifically Google Gemini — with a community-driven learning environment, LearnPath provides students with a personalised, accessible, and culturally relevant academic companion that adapts to their individual needs.

1.2 Chosen Sustainable Development Goal (SDG)

LearnPath aligns with United Nations Sustainable Development Goal 4: Quality Education — specifically Target 4.1 (free, equitable, and quality primary and secondary education) and Target 4.4 (increasing the number of youth with relevant skills for employment). The platform is further supported by Google.org's Knowledge, Skills & Learning focus area, which prioritises technology-driven solutions that break down barriers to learning in underserved communities.

1.3 Problem Statement

Students across the ASEAN region face a compounding set of challenges that hinder equitable academic achievement. Firstly, traditional tutoring is financially inaccessible for most low-to-middle-income families. Secondly, classroom learning is often one-size-fits-all, leaving students who learn differently or at a different pace behind. Thirdly, study resources are predominantly available in English, which alienates non-English-speaking students. Finally, students frequently lack the motivation and structure required for independent learning outside the classroom.

Without timely, affordable, and personalised support, these students risk falling further behind, perpetuating cycles of educational inequality.

1.4 Project Objectives

LearnPath was built with the following core objectives:

- Provide AI-generated, personalised learning roadmaps tailored to each student's academic goals, education level, and country.
- Offer an always-available AI tutor that delivers contextually relevant, curriculum-aligned tutoring in a conversational format.
- Support ASEAN language diversity through a fully internationalised interface covering English, Bahasa Malaysia, Tamil, Vietnamese, and Filipino.
- Foster a collaborative learning culture through community study groups, shared resources, and peer interaction.
- Motivate continued learning through gamification features including points, badges, streaks, and leaderboards.

1.5 What Makes LearnPath Different

Unlike generic educational platforms, LearnPath was engineered from the ground up for the ASEAN context. The AI Tutor dynamically adjusts its teaching style and cultural references based on the student's country, grade level, and education system. Roadmaps are not templated — they are generated on-demand by Gemini 2.0 Flash and stored in Firestore, meaning every student receives a unique study plan.

Furthermore, the internationalisation system supports five languages with a modular architecture ready for Khmer and Thai expansion. The platform is entirely free to use, removing financial barriers that typically exclude low-income students from quality academic support tools.

1.6 Project Vision

The long-term vision of LearnPath is to become the go-to AI learning companion for every secondary school student across Southeast Asia — a platform that understands who you are, where you come from, and where you want to go, then helps you get there one step at a time.

2.0 Review of Existing Solutions

The global edtech landscape is rich with solutions, yet most fail to adequately serve ASEAN students. A comparison of leading platforms reveals significant gaps that LearnPath is uniquely positioned to fill.

Platform	Strengths	Limitations	LearnPath Advantage
Khan Academy	Free, structured content	English-only, no AI personalisation	Multilingual + AI roadmaps
Duolingo	Gamified, engaging UX	Language-learning only	Covers all academic subjects
ChatGPT / Gemini	Powerful AI responses	No learning structure or progress tracking	Structured goals + progress + community
Brainly	Peer Q&A, regional presence	Passive content, no personalisation	Active AI tutoring + personalised plans

This analysis confirms that while individual components of LearnPath exist elsewhere, no single platform combines AI-personalised roadmaps, multilingual support, real-time tutoring, community features, and gamification specifically for ASEAN secondary school students.

3.0 Methodology

3.1 Technical Stack

LearnPath is built on a modern, scalable, serverless architecture designed for rapid development and low operational cost — both critical constraints for a social impact project.

Frontend

- React 19 + Vite 7: Component-based UI with hot module replacement for fast development.

- Lucide React: Consistent, accessible icon library across all UI components.
- Recharts: Data visualisation for progress tracking and leaderboards.
- Tailwind CSS (CDN): Utility-first styling for rapid, responsive UI development.

Backend (Firebase)

- Firebase Authentication: Supports Email/Password and Google OAuth sign-in.
- Cloud Firestore: NoSQL database storing users, goals, communities, posts, and sessions.
- Firebase Storage: Hosts student-uploaded study documents (PDFs, images) with categorised metadata.
- IndexedDB Persistence: Auth state is persisted via IndexedDB for reliable multi-tab and multi-browser sessions.

AI Integration

- Google Gemini 2.0 Flash: Powers both the AI learning roadmap generator and the real-time AI tutor.
- Structured Prompt Engineering: Enforces JSON output for roadmaps and Markdown-formatted responses for tutor chat.
- Context-Aware Tutoring: Each tutor request includes the student's education level, grade, country, and current learning goal.

3.2 Non-Technical Approach

Beyond the technical architecture, LearnPath was designed with a human-centred methodology that prioritises accessibility, motivation, and cultural relevance.

Internationalisation (i18n)

A custom i18n system was built using React Context API, supporting 200+ translation keys across five languages. Language preference persisted in localStorage, and the system includes an automatic fallback to English when a key is missing. The modular design allows new languages to be added by simply extending the translations file.

Gamification Design

The gamification system was designed using principles from behavioural psychology — specifically variable reward schedules and progress visibility. Students earn points for completing goals, maintaining login streaks, and engaging with the community. Badges are awarded dynamically at defined milestones (e.g., '7 Day Streak', 'First Goal Completed'), and a public leaderboard channels healthy competition.

Community & Collaboration

Study communities allow students to create or join subject-specific groups where they can share posts, upload categorised study resources, and schedule collaborative sessions. Firestore's real-time listeners (onSnapshot) ensure all activity is synchronised instantly across all connected users without page refreshes.

4.0 Results

4.1 Summary of Outcome

LearnPath successfully delivers a fully functional, AI-powered learning platform targeting ASEAN secondary school students. The prototype demonstrates end-to-end functionality across all core features: user authentication, AI roadmap generation, real-time tutoring, community collaboration, document management, and gamification.

4.2 Objectives Achieved

- AI Roadmap Generation: Gemini 2.0 Flash reliably produces 5-step structured learning plans saved to Firestore.
- AI Tutor: Context-aware tutoring with conversation history, personality modes, and mobile-optimised formatting.
- Multilingual Support: Five ASEAN languages fully implemented with persistent user preference.
- Community Features: Real-time study groups, post sharing, document uploads, and session scheduling.
- Gamification: Points, streaks, badges, and leaderboard fully integrated with Firestore.

4.3 Data and Feedback Gathered

During internal testing with a small group of secondary school students in Malaysia, key qualitative findings emerged. Students rated the AI tutor's explanations as clearer and more contextually relevant than generic AI tools, particularly appreciating references to Malaysian exam formats (SPM). The Bahasa Malaysia interface was cited as a major confidence booster for students less comfortable with English. The gamification system was noted as a strong motivator to return to the platform daily.

Technically, the Gemini API demonstrated high reliability with well-engineered prompts, producing consistently valid JSON for roadmaps. The main challenge encountered was occasional Markdown wrappers around JSON responses, resolved through a RegExp cleanup step in the geminiService.js module.

4.4 Next Steps for Validation (Recommended KPIs)

- Daily Active Users (DAU) and 7-day retention rate to measure engagement.
- Average roadmap completion rate per user as a measure of learning progress.
- Language distribution metrics to confirm multilingual adoption.
- AI tutor satisfaction rating (1-5 star post-session survey).
- Community post volume and document upload frequency as collaboration indicators.

5.0 Prototype Showcase

The LearnPath prototype comprises the following key user-facing screens and interactions:

Landing Page & Authentication

A visually engaging landing page introduces the platform's value proposition with a language switcher allowing users to select their preferred ASEAN language. Authentication supports both email/password registration and Google Sign-In, with Firebase handling session persistence.

Dashboard & Goal Creation

Upon login, users land on a personalised dashboard displaying active goals, progress statistics, current streak, and point balance. The goal creation dialog prompts for a learning objective, then

calls the Gemini API to generate a tailored 5-step roadmap displayed as an interactive progress tracker.

AI Tutor Interface

The tutor chat interface supports multiple AI personalities (General Tutor, Math Expert, etc.). Each message is sent with the student's full profile context, and responses are rendered in formatted Markdown optimised for mobile readability. Conversation history is maintained within the session for coherent multi-turn dialogue.

Community Hub

The community section allows students to browse and join study groups by subject. Within each group, members can post updates, upload categorised study materials (Notes, Assignments, Past Papers), and schedule or join study sessions. All interactions are synchronised in real time via Firestore listeners.

Gamification & Leaderboard

A dedicated achievements panel displays earned badges, current level, total points, and login streak. The public leaderboard ranks all users by point total, encouraging healthy competition and sustained platform engagement.

6.0 AI Acknowledgement

6.1 AI Tools Used

- Google Gemini 2.0 Flash — Core AI engine for the platform's tutoring and roadmap features.
- Claude (Anthropic) — Used as a development assistant during the build process.
- GitHub Copilot — Assisted with boilerplate code generation and auto-completion.

6.2 How We Utilised Each Tool

Google Gemini 2.0 Flash is embedded directly into the LearnPath platform via the Gemini REST API. It powers two core features: the learning roadmap generator (structured JSON output via

enforced prompt templates) and the AI tutor (Markdown-formatted conversational responses with dynamic context injection). Extensive prompt engineering was conducted to ensure the model produced consistently formatted, educationally appropriate outputs.

Claude was used during development as a coding assistant and documentation aid — helping debug React component issues, optimise Firestore query patterns, and review code structure. It was not used to generate any end-user-facing content within the platform.

GitHub Copilot accelerated repetitive coding tasks such as Firebase CRUD operations and Tailwind styling, reducing development time without compromising code quality.

6.3 Data & Consent

LearnPath does not collect or transmit any personally identifiable student data to third-party AI providers beyond what is necessary for the tutoring interaction. Tutor messages are sent to the Gemini API with minimal contextual metadata (education level, grade, country — no names or IDs). Firebase Authentication and Firestore operate under Google's standard data processing terms. No biometric or sensitive personal data is collected at any point.

7.0 Conclusion

LearnPath demonstrates that a small, purpose-built team can leverage modern AI and cloud infrastructure to create a meaningful, scalable solution to educational inequality in the ASEAN region. By combining Gemini-powered personalisation, multilingual accessibility, community collaboration, and motivational gamification into a single free platform, LearnPath directly advances SDG 4 and Google.org's Knowledge, Skills & Learning agenda.

The prototype validates the core technical architecture and confirms genuine demand for culturally relevant, AI-assisted learning tools among ASEAN secondary school students. The platform is ready for user testing at scale and positioned for rapid iteration based on real-world feedback.

7.1 Future Improvements

- Interactive AI Quizzes: Gemini-generated micro-quizzes derived from uploaded study documents to test knowledge dynamically.
- Automated Document Analysis: Auto-generate summaries and flashcards from uploaded PDFs using Gemini's document understanding capabilities.
- Mobile Application: Port to React Native for a dedicated iOS and Android experience with native push notifications and offline support.
- Khmer & Thai Language Support: Extend the i18n system to include Cambodian and Thai, expanding ASEAN reach to over 80 million additional students.
- Video/Audio Study Sessions: Integrate WebRTC or Agora for live peer-to-peer video study sessions directly within the platform.
- Advanced Gamification: Introduce seasonal leaderboards, customisable avatars, and an in-app economy tied to study hours.
- RTL Language Support: Enhance the i18n architecture to support right-to-left languages such as Arabic and Urdu for potential Middle East expansion.
- Offline Mode: Implement service workers and local caching to allow students with intermittent internet access to continue learning.

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